Electrocardiogram Signal–Based Cardiac Abnormality Detection

Dataset

MIT-BIH Arrhythmia Database

The dataset contains 48 ECG recordings from different patients.

Each recording is approximately 30 minutes long and sampled at 360 Hz.

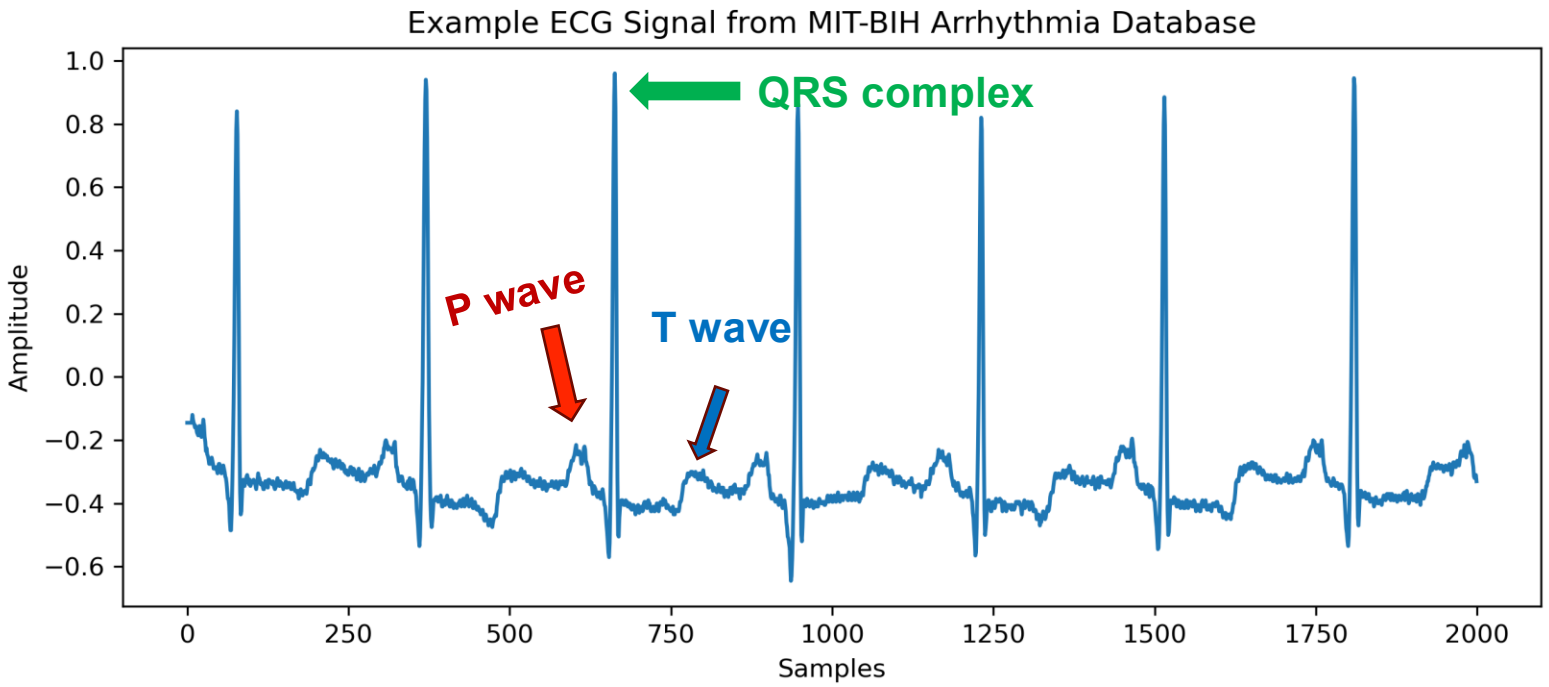
Data Characteristics:

An ECG signal typically includes the following components:

P wave – represents atrial depolarization

QRS complex – represents ventricular depolarization and indicates a heartbeat

T wave – represents ventricular repolarization



Import dataset

The ECG data were imported using the **WFDB Python library**.

Each record consists of three files:

- dat – ECG signal
- hea – signal metadata
- atr – heartbeat annotations

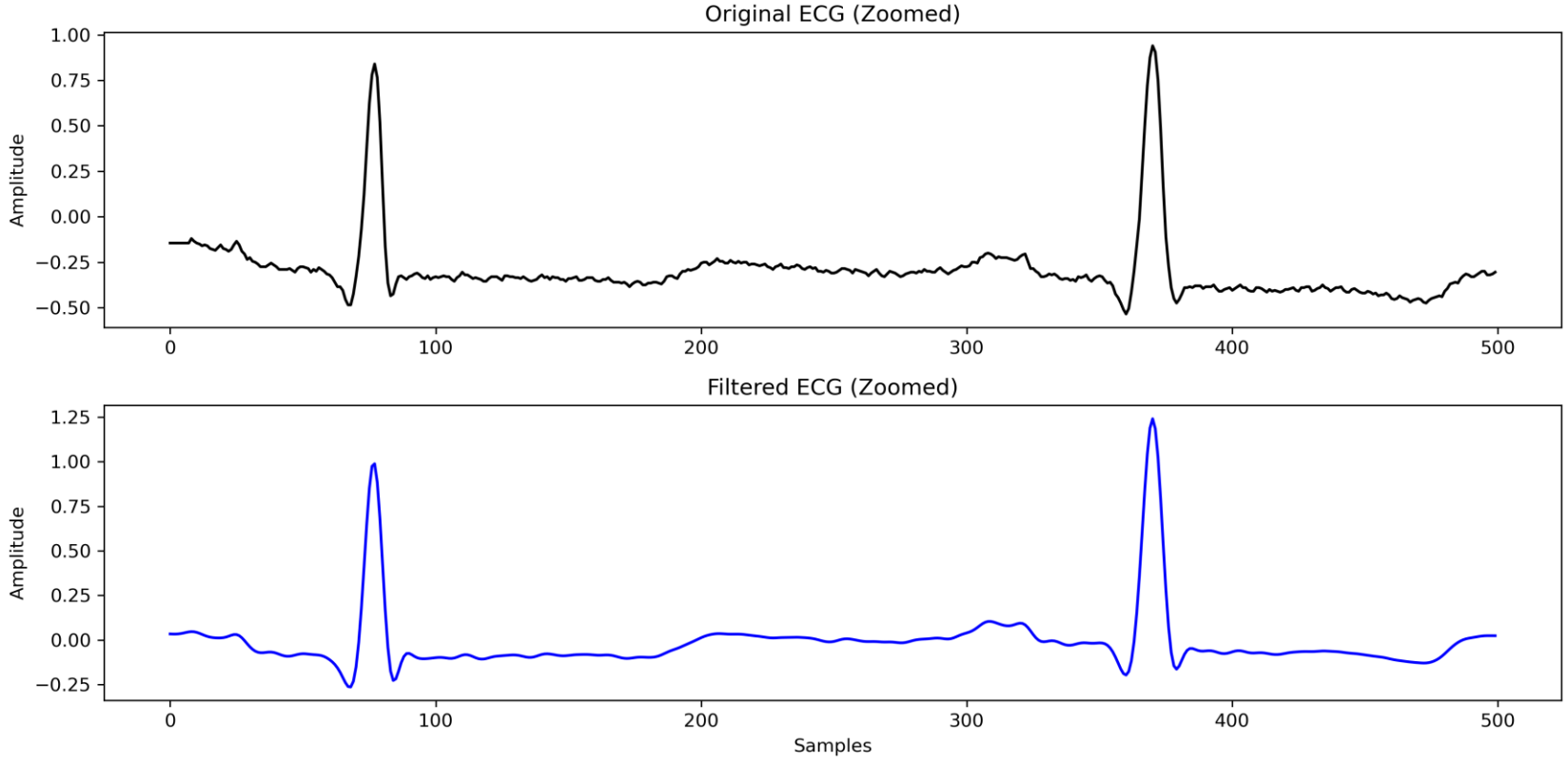
Python

```python
import wfdb

record = wfdb.rdrecord('mitdb/100')
annotation = wfdb.rdann('mitdb/100','atr')

signal = record.p_signal[:,0]
```

Data Processing



Noise in ECG signals

- Baseline wander
- Powerline interference
- High-frequency noise

Filtering Method

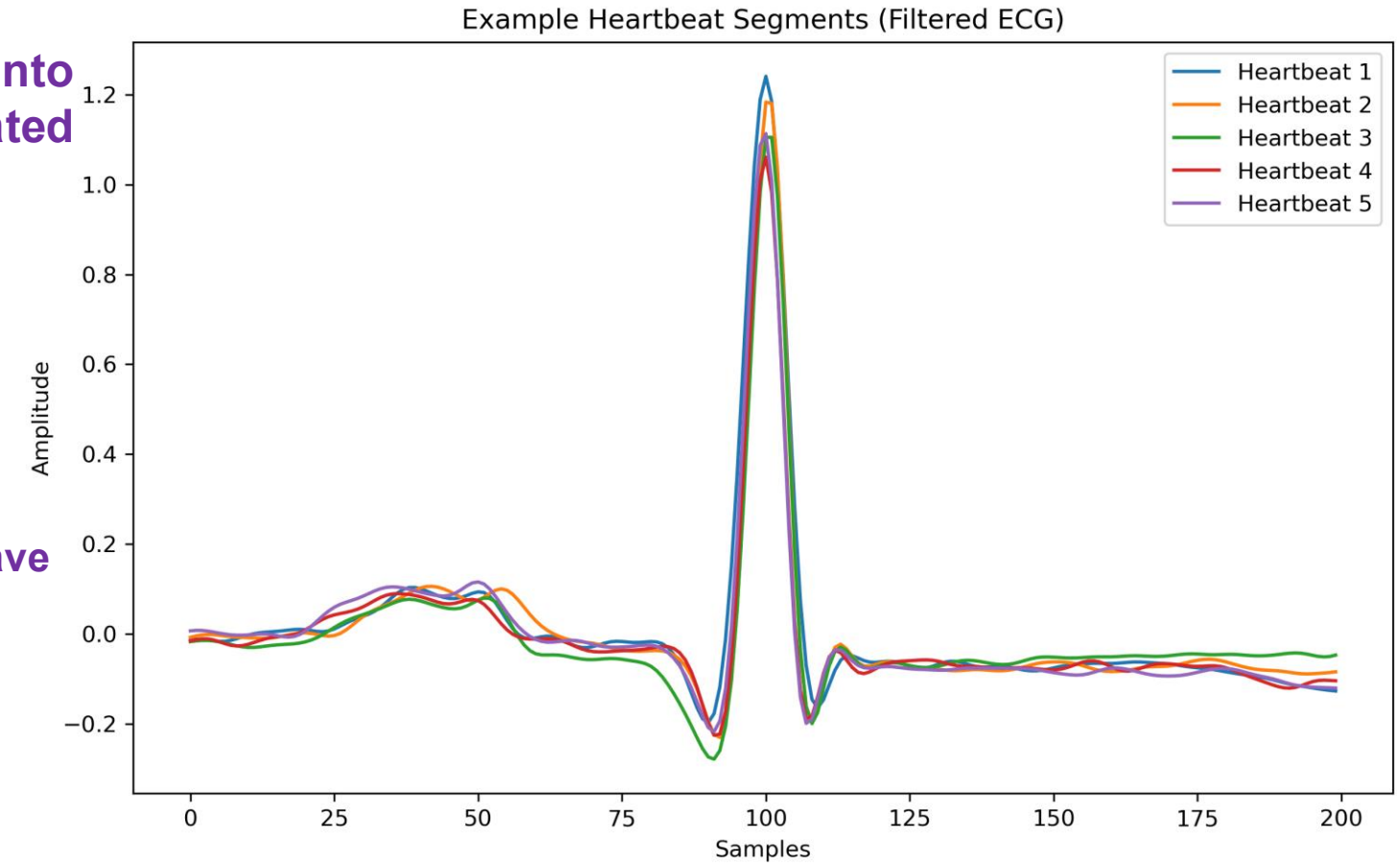
- Bandpass filter (0.5–40 Hz)
- Remove noise
- Preserve ECG features

Heartbeat Segmentation

The filtered ECG signal is segmented into individual heartbeats using the annotated R-peak positions.

Each heartbeat segment:

- centered around the R-peak
- contains 200 samples
- includes P wave, QRS complex, and T wave



Normalization

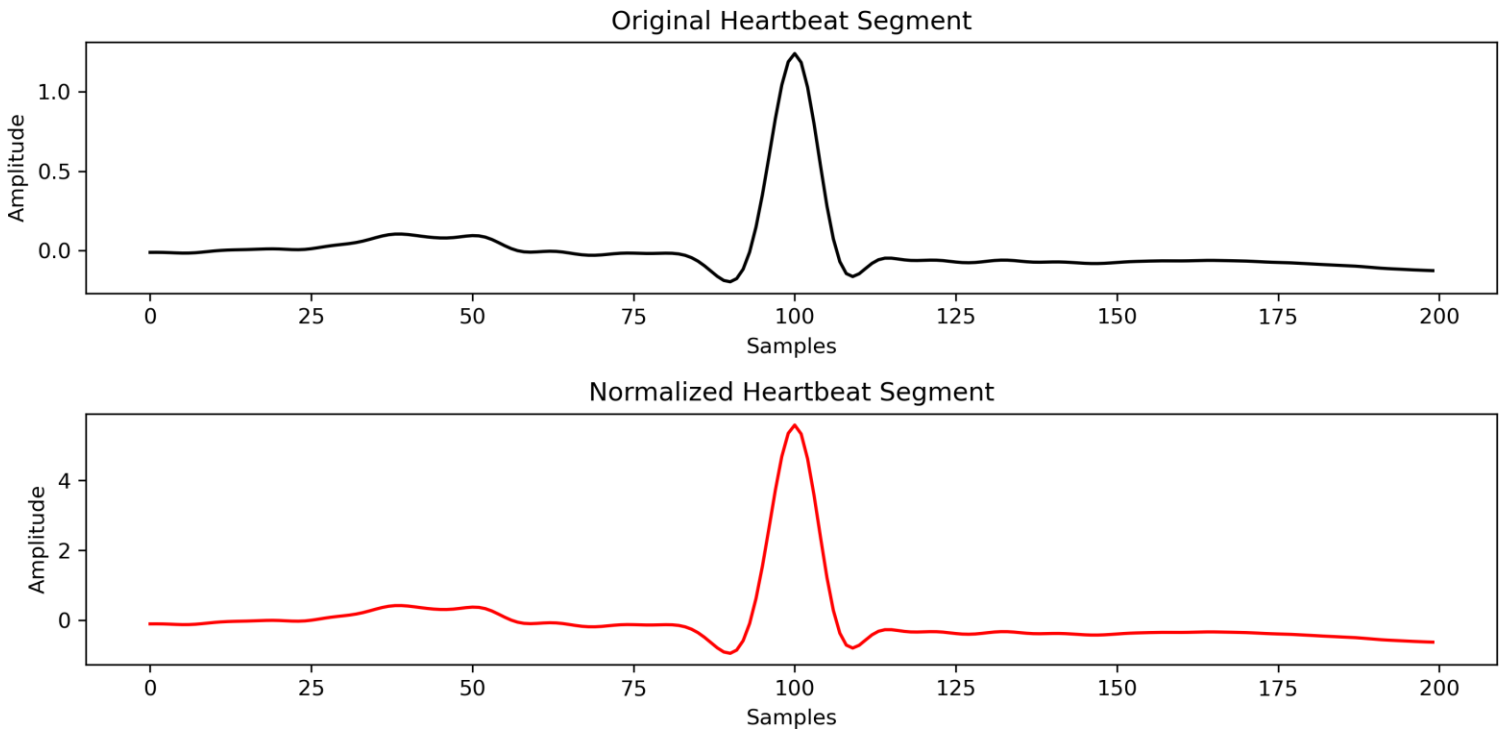
Normalization transforms the signal using:

$$x' = \frac{x - \mu}{\sigma}$$

μ : mean of the signal
σ : standard deviation

This operation:

- **centers the signal around zero**
- **rescales the amplitude**
- **preserves the waveform structure**



Model Architecture

Raw ECG Signal	→	Beat Windowing	→	Standard Scaler	→	PCA	→	SVM	→	Normal / Abnormal
WFDB format .dat / .hea / .atr		180 samples per beat		Normalize amplitude		180 → 30 features		RBF kernel C=10, balanced		Binary classification

PCA — Principal Component Analysis

Purpose: Reduce 180 raw ECG time-steps → 30 key components

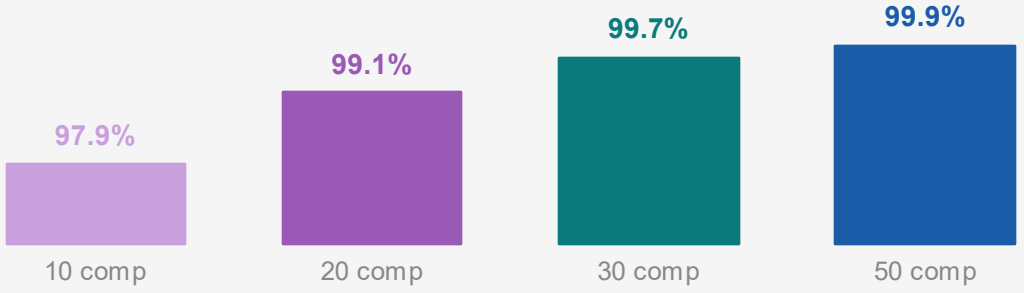
Variance kept: 99.7% (almost no information lost)

How: Finds patterns that vary most across all 112,579 beats

Component 1 → R-peak height (main heartbeat spike)

Component 2 → QRS complex shape

Component 3+ → P-wave, T-wave patterns



Variance Retained vs. Components

SVM — Support Vector Machine

Purpose: Draw the optimal boundary between Normal & Abnormal beats

Kernel: RBF (Radial Basis Function) — handles non-linear boundaries

Key parameters:

Initial SVM	Improved SVM
`C=1.0` `No class weight`	`C=10.0` `class_weight='balanced'`
Abnormal Recall: **76%**	Abnormal Recall: **94%**

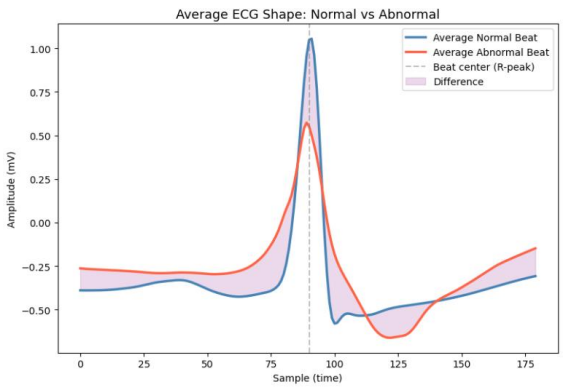
Why it worked: Dataset has 4× more normal beats (90k) than abnormal (22k). Balancing forced the model to treat both classes equally.

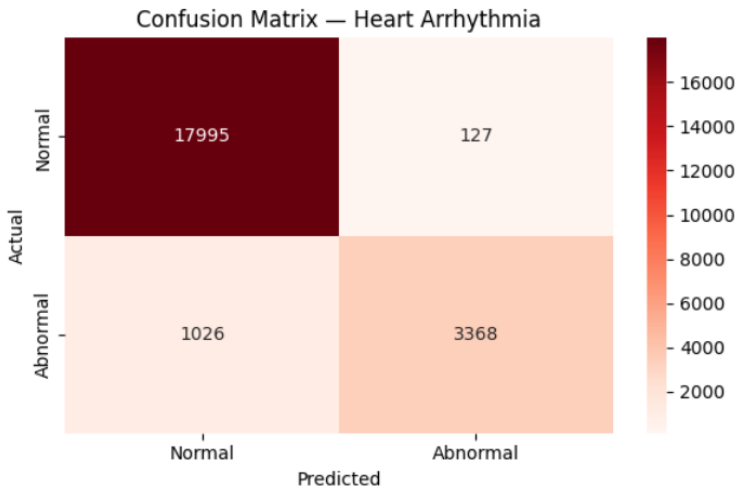
Model Results

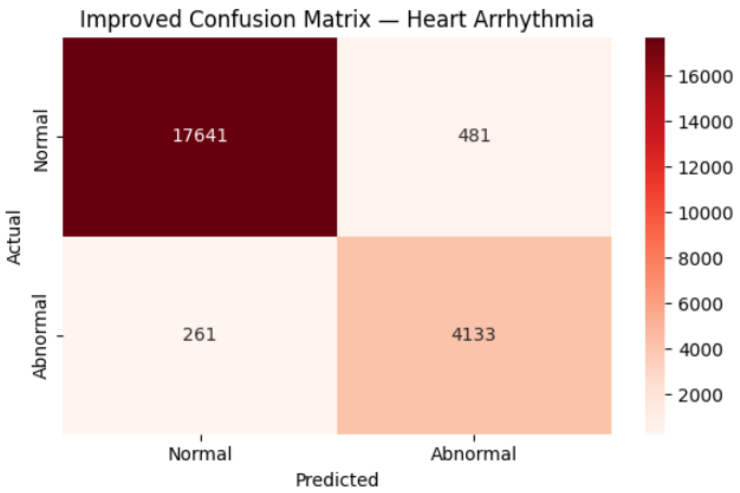
97%	94%	89%	92%
Overall Accuracy	Abnormal Recall	Abnormal Precision	F1-Score

Before vs. After Class Balancing

Metric	Initial Model	Improved Model
Accuracy	95%	97% ↑
Abnormal Recall	76%	94% ↑
Missed Beats	1,050	253 ↓







Key Findings

- Class balancing reduced missed patients from 1,050 → 253 doing this
- PCA retained 99.7% variance using only 30 components Where 10 components changed the variance to 97.9% and 50 components changed the variance to 99.9%
- SVM with RBF kernel C=10 proved most effective rather than increasing the components or samples in SVM as the other 2 options would slow the speed of model
- A defining component that stands out is the average amplitude of the normal heartbeats vs abnormal heartbeats it is seen abnormal beats have an average of .6mV and normal heartbeats averaging at 1mV.
- *Next step: 1D CNN for comparison*